

Letter of Support:

Keyrack Project

NGI Zero Commons Fund Application

From: Borislav Borislavov, CTO, OpsHero

Date: 29.01.2026

Re: Technical Support for Keyrack — Open Source Key Lifecycle Coordination

To the NGI Zero Review Committee,

I am writing to provide technical endorsement for the Keyrack project's application to the NGI Zero Commons Fund.

As Chief Technical Office at OpsHero, I lead integration work across multiple cloud providers and on-premises infrastructure for our clients. Key management is a consistent pain point, and the gap Keyrack addresses is one I encounter regularly.

The technical problem is real. We deploy and integrate systems using HashiCorp Vault, AWS KMS, Azure Key Vault, and various HSMs. Each provides solid cryptographic operations—key generation, encryption, decryption. None provides lifecycle coordination. When a customer asks "what happens if we need to rotate this KEK," the honest answer is: we would need to manually trace every system that depends on it, coordinate re-wrapping across services, and hope we didn't miss anything. There is no tooling for this. We build custom scripts for each engagement, and they are brittle. With regards to post-compromise DEK rotation - the situation is even more dire, most teams just assume "nothing can be done" and do not plan or mitigate for this scenario.

Dependency tracking does not exist. I have evaluated commercial key management solutions for clients. They offer scheduled rotation, compliance dashboards, policy engines. What they do not offer is the ability to query "what data or downstream keys depend on key X." This capability is assumed to exist somewhere in the stack, but it does not. The Keyrack proposal correctly identifies this as the missing layer.

Provider-agnostic coordination is valuable. Our clients use heterogeneous infrastructure—some workloads in AWS, some in Azure, some on-premises with Thales HSMs. A coordination layer that abstracts across providers, tracking hierarchy and dependencies regardless of where keys physically reside, would significantly reduce integration complexity. The architecture described in the Keyrack proposal - sitting above cryptographic backends, delegating operations while managing lifecycle - is the right approach.

The technical design is sound. I have reviewed the proposal's technical approach: attribute-based key identity, routing rules for hierarchy definition, and the tiered security model (KMIP backends with no material exposure, Transit as documented degradation). These are sensible architectural choices. The separation between Keyrack coordinating and applications executing re-encryption correctly places domain-specific knowledge where it belongs.

Open source infrastructure benefits the ecosystem. Key lifecycle coordination should not be proprietary. It is foundational infrastructure that every secure system needs. An open source solution - particularly one designed for self-hosting and sovereignty—benefits the entire European technology ecosystem.

I believe Keyrack addresses a genuine technical gap and is well-designed to solve it. The project merits NGI Zero support.

Sincerely,

Borislav Borislavov

CTO, OpsHero.com

b.borislavov@opshero.com, opshero.com, bul Simeonovsko Shose 33, office 1, floor 1, Sofia, Bulgaria

